INCH-POUND
MIL-DTL-24643/7F
1 October 2009
SUPERSEDING
MIL-DTL-24643/7E
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO +90 °C, 600 VOLTS, TYPE LSMHOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, coated or uncoated, 16 AWG, Class K stranding.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Minimum average wall thickness of 0.020 inch. Standard identification code applied by Method 1 or Method 3. If identification Method 1 is used, insulation color shall be black.
Fourth	-	The specified number of conductors (see table I) shall be cabled together as required for the completed cable to meet the twisting and bending endurance tests. The outer layer shall be a left-hand lay. The inner layer(s) lay direction at the manufacturer's option. Fillers may be used as necessary to form a firm well-rounded assembly. Cabling sequence to be consecutive, starting with no. 1, from the center outward. A single conductor shall not be used in the center for the cable core.
Fifth	-	An optional binder.
Sixth	-	Cross-linked polyolefin jacket. Minimum average wall thickness of 0.060 inch

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TABLE I. Details.

Military part no. M24643/7	Type and size	Number of conductors	Overall diameter	
			Min. (inch)	Max. (inch)
-01UN	LSMHOF-7	7	0.465	0.500
-02UN	LSMHOF-10	10	0.540	0.585
-03UN	LSMHOF-14	14	0.585	0.635
-04UN	LSMHOF-19	19	0.650	0.705
-05UN	LSMHOF-24	24	0.735	0.795
-06UN	LSMHOF-30	30	0.775	0.835
-07UN	LSMHOF-37	37	0.855	0.925
-08UN	LSMHOF-44	44	0.925	1.000
-09UN	LSMHOF-61	61	1.100	1.175

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	4.59
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Twisting and bending endurance (see table I for roller size)	
At 75 °C (cycles, min.)	6000
At minus 30±2 °C (cycles, min.)	2000
Physicals (unaged)	
Insulation (extruded)	

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Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) air oven	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation, Method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2

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Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
All sizes	1500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2008-011)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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